MILITARY SPECIFICATION

TEST SET, ADAPTER GROUP, OF-31(V)1/APM

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for shop use and shall provide signals and measuring devices necessary for the alignment and calibration of the AN/APN-194(V) Electronic Altimeter Set when used in conjunction with the AN/APM-347 or TS-1979A Test Sets.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-T-21200	Test Equipment for use with Electronic and Electrical Equipment, General Specification for
MIL-T-18303	Test Procedure; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration; in so doing the following shall govern:

(a) Parts and materials requirements shall be in accordance with MIL-T-21200. Parts selected for use which are considered nonstandard per MIL-T-21200 shall be selected to meet the design requirements. Such parts shall only be used when a standard MIL equivalent part is not available or when delivery time of the standard part is beyond the time limit required to meet the contract delivery schedule.

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(b) Semiconductor Devices - Semiconductors shall be chosen and applied in accordance with MIL-STD-454 Requirement 30.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained in accordance with MIL-T-21200.

3.2 Design and Construction - The design and construction of the equipment shall be in accordance with MIL-T-21200 Class 3 equipment.

3.2.1 Weight - The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 47 pounds.

3.2.2 Reliability

3.2.2.1 Operating Life - The equipment shall have a minimum total operating life of 45,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment, and the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.2.2.2 Reliability in Meantime Between Failure - The calculated mean (operating) time between failure shall be 5,000 hours.

3.2.3 Cabling and Connections - The equipment shall provide for the use of cables and connectors in accordance with MIL-T-21200.

3.2.3.1 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with MIL-W-5088. The external wiring may be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements of MIL-T-21200.

3.2.3.1.1 Wires - Individual wires shall have white jackets. Wire identification shall not be a requirement.

3.2.4 Control Panels - All rack mounted control panels shall conform to the applicable requirements as specified in MIL-T-21200.

3.2.5 Interchangeability - The equipment shall meet the applicable interchangeability requirements as specified in MIL-T-21200.

3.2.6 Interference Control - The equipment shall be designed to meet the radio frequency interference control requirements of MIL-T-21200 minimizing the vulnerability of the equipment to Radio Frequency Interference.

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3.2.7 Maintenance Provisions and Field Testing - Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance. Provisions for maintenance shall be as specified in MIL-T-21200.

3.2.8 Nomenclature and Identification - Nomenclature assignment and identification approval for equipment identification shall be in accordance with MIL-N-18307.

3.2.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ± 5°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input pwr. volt.	115 ± 6.0 VAC, 400 ± 20Hz +27 VDC ± 2 VDC

3.2.10 Service Conditions - The equipment shall operate satisfactorily when subjected to the environmental service conditions in accordance with MIL-T-21200 for Class 3 equipment.

3.2.11 Warm-Up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 5 minutes under standard conditions and 10 minutes at extreme service conditions.

3.2.12 Primary Input Power - The equipment shall give the specified performance from the following power source with characteristics as specified in MIL-T-21200 having limits as specified herein. The power required shall not exceed:

(a) 115 Volts, ± 10%, 400 ± 20Hz, Single Pnase, 300VA
(b) +27 VDC, ± 2 VDC, 50 Watts

3.3 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.1 Operation - The Test Adapter, when used in conjunction with the AN/APM-347 or the TS-1979A Test Sets, shall provide for test of the AN/APN-194(V) Altimeter at the bench or hangar level.

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3.3.1.1 Test capabilities shall be provided as follows:

(a) The Test Adapter will have all vast (Versatile Automatic Shop Test) test points available for monitoring system operation.

(b) The Test Adapter shall provide the necessary interface wiring to enable the TS-1979A or the AN/APM-347 to properly test and align an AN/APN-194 Electronic Altimeter.

(c) The Test Adapter shall provide a manual override enable and adjustment control to provide adjustment of the altitude output signals to any value between 0 and 5000 ft.

(d) The Test Adapter shall provide switches to test the following parameters.

1. Manual track/enable (NO TRK-NORM-TRK)
2. Search enable (SRC Assist)
3. Self test enable (Self Test)
4. Sensitivity Range control (Range)
5. Radar aural warning signal (RAWS Enable)
6. Flag circuit (Flag Cont).

(e) Provide at least 20db isolation between the Altimeter test set and the Altimeter under test.

3.4 Detail Requirements

3.4.1 The Test Set Adapter, AN/APM-347 shall meet the following requirements:

3.4.1.1 Detailed Requirements

A. The adapter shall have provisions to initiate or control the following radar altimeter functions:

1. A manual override enable (slew off) and adjustment control (Range) to provide adjustment of the altitude output signals to any value between 0 and 5000 feet (0V to +28V).

2. Manual Track/Enable (NO TRK-NORM-TRK) -15V to +15V.

3. Search Enable (SRC Assist) 0V to +15V.

4. Sensitive Range Control (Range) 0V to -15V.

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5. Self Test Enable OPEN or CLOSED contacts.

6. Radar Aural Warning Signal (RAWS) OPEN or CLOSED contacts.

7. Flag circuit (Flag Cont) OPEN or CLOSED contacts.

8. Blanking Pulse Enable (BLKG PLS IN-NORM) +15VDC or 91Ω termination.

9. Forced Short Pulse OPEN or CLOSED contacts to gnd.

B. Indicators shall be provided to display the reliability signal.

C. Adapter cables for connecting the test adapter to the TS-1979A and/or the APM-347 shall be provided. Cables shall also be required for connection of the the test adapter to the AN/APN-194.

D. The Test Adapter, Epsco Part No. 2019 shall be capable of accepting a digital test panel, Epsco Part No. 2019-0001. Power required to operate the digital test panel and other interconnecting wiring shall be provided by the basic test adapter.

E. R.F. isolation between the Altimeter Test Set and the Altimeter under test is required.

3.4.1.2 Form Factor - The system test set shall be approximately (13 5/32) inches deep,(19 21/32) inches wide and (14 3/32) inches high. The case assembly being (7 5/32) inches high and the cover assembly being (6 15/16) inches high.

3.4.1.3 Weight - The weight of the system test set shall not exceed 47 pounds.

3.4.1.4 Cover and Case Assembly - The equipment shall be housed in a combination case, composed of two sections, the Cover Assembly and the Case Assembly. It shall have the capability of being separated, and be in accordance with the requirements of MIL-T-21200.

3.4.1.4.1 Cover Assembly - The cover assembly shall have a hinged inner lid and shall be used to store and transport eleven cable assemblies. Cover indentations to match the feet on the bottom of the case to facilitate stacking need not be required.

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3.4.1.4.2 Case Assembly - The case assembly shall have a removable blank panel assembly for the digital test panel PL 2019-0001.

3.4.1.4.3 The cover assembly and the case assembly shall each have two handles.

3.4.1.4.4 The case assembly shall have two sets of feet, one set on the bottom and one set on an end as provision for resting the case on a bench.

3.4.1.4.5 Combination Case latches shall be in accordance with MIL-T-21200.

3.4.1.5 Controls - The equipment controls, their functions and location are as follows:

Control Name	Function	Location
Flag Control	Indicator Test	Front Panel
RAWS Enable	+ 28VDC to Ind.	Front Panel
Self Test	Switch Closure	Front Panel
SRC Assist	Switch Closure	Front Panel
PWR	Turn-On AC Power	Front Panel
NO TRK-NORM-TRK	± 15VDC to RT	Front Panel
Slew - OFF	Enables Range Control	Front Panel
Range	Test Voltage(Range)	Front Panel
BLKG PLS IN	+ 15VDC/91Ω termin.to RT	Front Panel
IND - SYS	Applies Load to Ind.	Front Panel
Forced Short Pulse	Switch Closure to Gnd.	Front Panel
+ 15V Adjust	Power Supply Adj.	Internal
- 15V Adjust	Power Supply Adj.	Internal

3.4.1.6 Test Set Indicator

Indicator Name	Function	Location
Rel + 4V	Display Rel Sig	Front Panel
115VAC	Indicates 115VAC On	Front Panel

3.4.1.7 Electrical Connections

Ref. Designation	Function	Receptacle Type
J1	To Indicator	MS 3112E18-32S
J2	To R.T. Under Test	MS3112E18-32SW
J3	To R.T. Under Test	MS3112E12-10SW
J4	To R.T. Under Test	MS3112E12-10S
J5	To Test Set	MS3112E14-19S
J6	To Test Set	MS3112E14-19SX
J7	To Test Set	M39012/28-0001
J8	To R.T. Under Test	M39012/28-0001

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Ref. Designation	Function	Receptacle Type
J9	Input Power	MS3112E14-5P
J10	To R.T. Under Test	M39012/28-0001
J11	To R.T. Under Test	M39012/28-0001
J12	Power & Signals to Digital Panel	MS3114-E14-15S

3.5 Associated Equipment - This equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment.

(a) Epsco Panel Digital Test PN 2019-0001.
(b) TS-1979A or APM-347 Test Sets.

4. QUALITY ASSURANCE

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(a) Preproduction Tests
(b) Production Tests
(c) Acceptance Tests

4.1.1.1 Preproduction Tests - When specified in the contract, preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. The tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.1.1.1.1 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract, within the limits specified in MIL-T-21200, first article tests may include environmental tests in accordance with MIL-T-5422 and interference test in accordance with MIL-STD-462 to the MIL-STD-461 requirements.

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4.1.1.1.1.1 Preproduction Approval - Approval of the test sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipment shall be delivered prior to the approval of the test sample. Prefabrication of production equipment prior to the approval of the test sample is at the contractor's own risk. The approved test sample may be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance.

4.1.1.1.2 Production Equipment - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make the necessary corrections as approved by the procuring activity.

4.1.1.2 Production Tests - Each production unit shall undergo tests to insure compliance with the requirements of this specification. This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved test sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.1.1.2.1 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.1.1.2.2 Reconditioning of Production Test Sample - On completion of the production test, the equipment may be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor may resubmit the equipment for acceptance.

4.1.1.3 Acceptance Tests - The supplier shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Suppliers not having testing facilities satisfactory to the procuring activity shall engage the services of a commercial testing laboratory acceptable to the procuring activity. The supplier shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the supplier. Acceptance tests shall consist of individual unit tests. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product.

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4.1.1.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability specified herein. As a minimum, each equipment accepted shall have passed the following tests:

(a) Examination of Product
(b) Operational Test

4.1.1.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.1.1.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation. Approval of the operational test procedure must be obtained prior to testing.

4.1.1.4 Reliability Assurance Tests - Reliability assurance tests for this equipment are not a requirement of the contract.

4.1.2 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector, to modify or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Test specifications shall be prepared in accordance with MIL-T-18303.

4.1.3 Presubmission Testing - No item, part or complete equipment shall be submitted until it has been previously tested and inspected by the supplier and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.1.4 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejections and the action taken to correct the defects found in the original shall be furnished.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract.

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6. NOTES

6.1 Intended Use - This equipment is intended to provide the test capability required for fault isolation, repair and adjustment of the AN/APN 194(V) Altimeter when used in conjunction with the TS-1979A or APM-347 Test Set down to the module level.

6.2 Test Values - Normal and limiting values of performance data shall be determined at an input voltage of 115 ± 6.0VAC and + 27 VDC ± 2VDC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than specified in MIL-T-21200, it is desired their use be investigated.

6.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specification.

(b) This Specification - This Specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 Definition - The following non-military definition shall apply throughout this specification.

6.5.1 Supplier - The facility fabricating, assembling, acceptance testing and furnishing the equipment in accordance with the contract specifications.

Preparing activity:
Navy - AS
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DOCUMENT IDENTIFIER AND TITLE MIL-T- 81961(AS)
TEST SET ADAPTER, GROUP OF - 31(V)1/APM

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

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